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Transition to a Fully Funded Pension System

in an Aging Russia

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Contents

List of Acronyms

PAYGO Pay-As-You-Go

FF Fully Funded

Rosstat Federal State Statistics Service

OLG Overlaping Generations model

SFR Social Fund of Russia

NSPFs Non-State Pension Funds

UN United Nations

LBFGS Limited-memory Broyden–Fletcher–Goldfarb–Shanno

1 Introduction

The shift from a Pay-As-You-Go (PAYGO) state-funded pension scheme to a Fully Funded (FF) system, along with its implications, is a topic of discussion in numerous countries worldwide. This is because the PAYGO pension model, which currently prevails globally, has become outdated and no longer serves its intended purpose effectively.

The solidarity pension system, where current workers fund the pensions of current retirees, was introduced in most countries early in the 20th century. It became the predominant model during the "Baby Boom" period, which also shaped its key characteristics. In a PAYGO system, pension benefits for retirees rely entirely on taxes collected from the active workforce. This system was designed at a time when there were many workers relative to retirees, making it feasible to finance pensions through current contributions.

Today, the situation has reversed: the Old-Age Dependency Ratio – the number of people over 65 compared to those aged 18 to 65—is much higher than it was in the mid-20th century. According to forecasts by the Federal State Statistics Service (Rosstat), this ratio in the Russian Federation is expected to reach 50% or more by 2050 [1]. Under these circumstances, maintaining the PAYGO pension system in its current form is no longer feasible. The shrinking workforce will not generate enough funds to support the growing number of retirees. As a result, it becomes necessary either to reduce pension benefits, raise the retirement age, and significantly increase taxes, or to shift towards a fundamentally different pension system altogether.

A Fully Funded pension system is one in which there are no state pensions or pension taxes; instead, individuals are responsible for financing their own retirement through savings they have accumulated over their working lives. In this system, each person's pension depends on the funds they have personally set aside, rather than on transfers from the current workforce.

The focus of this paper is to examine the key factors that change as a result of transitioning from a public PAYGO pension system to a private funded system. Specifically, the study analyses how production, consumption, savings, and labour supply are affected. To do this, a general equilibrium Overlaping Generations model (OLG), tailored to the Russian economy, is developed. This model incorporates endogenous labour supply decisions, perfect foresight by individuals, realistic demographic transition dynamics, and a government tax sector. Building on this foundation, future research will explore how these factors evolve under different scenarios of moving from a PAYGO to a funded pension system.

Russia, in reality, already began its transition to a Fully Funded pension system

[1] The population of the Russian Federation by gender and age. Rosstat, 2022. Statistical bulletin.

in 2002. However, this new system quickly proved unsustainable and faced widespread criticism for several reasons: citizens had little control over their savings, the system was overly complex and opaque for workers, and the management of the Social Fund of Russia (SFR) was poor (Bekrenev & Arnautova, 2016). Over more than a decade, various proposals aimed at improving and expanding the system were put forward (Ahuja, Yermo, & McTaggart, 2013). Despite these efforts, the system effectively ceased to operate in 2014. This outcome reflects the challenges Russia faced in implementing pension reforms, including administrative weaknesses and political struggles that hindered the system's success.

In 2014, a law was enacted that eliminated the deduction of 6% out of the 22% insurance contributions from employees' salaries to private funded pension funds. This decision was officially justified by the claimed inefficiency of these private savings funds. Regional presidents of the SFR noted that in the 20 largest Non-State Pension Funds (NSPFs), which held over 70% of pension savings, average deposit growth ranged from only 2% to 8.3% annually, while inflation averaged 9.65% per year, effectively eroding returns [2].

However, this rationale is debatable since some pension funds were indeed profitable. Research by (Mischenko & Leonidova, 2018) showed that from 2009 to 2015, many NSPFs outperformed inflation, with a few showing significant gains. Among the 11 largest NSPFs, however, only four exceeded inflation, a result influenced by the high inflation rates during that period. In 2016, when inflation was low, all NSPFs achieved returns above consumer price growth. Overall, from 2009 to 2015, NSPFs yielded an average return of 76.6%, close to the 79.1% accumulated inflation, suggesting that under stable inflation conditions, conservative financial instruments can protect savings against inflation.

The 2014 reform was essentially a short-term measure to address budgetary pressures on the Pension Fund by redirecting funds away from the funded component back to current pension payments. It reduced the mandatory funded contribution from 6% to 2%, shifting the remaining 4% to cover immediate pension costs. This rollback was criticized for undermining the long-term sustainability of the pension system and for being a politically convenient but short-sighted solution that avoided raising the retirement age or implementing deeper reforms.

The Russian experience with partially funded pension systems cannot be regarded as a genuine or serious effort to transition fully from a PAYGO to a funded pension system. The reforms initiated since 2002, including the introduction of a mandatory funded pillar, faced significant administrative, financial, and political challenges that ultimately led to the system's rollback and effective cessation by 2014.

[2] https://sfr.gov.ru/branches/alania/news/interviev/~2014/08/19/404; https://sfr.gov.ru/branches/komi/news/interviev/~2014/08/08/126

Consequently, the question of achieving a realistic and viable complete shift to a fully funded pension system remains critically important and unresolved for Russia.

There is extensive literature in foreign journals that employs mathematical modelling of economies undergoing pension system transitions using OLG models. These works analyse the dynamic effects of shifting from PAYGO to funded pension systems by capturing key economic variables such as labour supply, savings, capital accumulation, and demographic changes within a rigorous general equilibrium framework. The theoretical foundations of OLG modelling in pension economics trace back to the seminal contributions of (Samuelson, 1958), whose work established the framework for analysing intergenerational transfers and dynamic efficiency in pension systems. Contemporary research has significantly expanded this foundation, incorporating stochastic elements, heterogeneous agents, and complex institutional arrangements to better capture real-world pension system dynamics (Barbie, Hagedorn, & Kaul, 2000).

The transition from PAYGO to funded pension systems represents one of the most extensively studied applications of OLG modelling. Research consistently identifies the fundamental challenge of transition costs, where current workers must simultaneously support existing retirees while building their own retirement accounts (Wang, Xu, Wang, & Zhai, 2004). If the government, however, decides to lower taxes on the workers to ease their saving for their old age, it will also face a challenge of a tax burden. The study (Makarski, Hagemejer, & Tyrowicz, 2017) raises the question of the most effective compensation for the tax gap after the transition to a funded system. Authors find that fiscal closure might be responsible for up to 30% of the overall welfare effect that pension reform has on the consumption of the transition-period cohorts. They show that a clever closure of a budget burden, involving both public debt financing and the taxation, gives the best results, redistributing the costs of transition fairly across generations.

The papers (Andersen, Bhattacharya, & Liu, 2021); (Andersen & Bhattacharya, 2017) consider the dynamic and Pareto-efficiency of PAYGO and Funded systems. The issue of efficiency on the example of the US economy is also discussed in (Nishiyama & Smetters, 2007). These papers come to the established conclusion that the Fully Funded pension system is more Pareto-efficient than the PAYGO. However, due to the challenges of the transition to the FF, in most cases the funded system becomes undesirable.

This quantitative part of this research (i.e. the results analysis) *will be* based on the approach established in (Lin, Tanaka, & Wu, 2021) and (Song, Storesletten, Wang, & Zilibotti, 2015). In these papers the transition dynamics on individual and aggregate levels is modelled and described in the most in-depth way. Authors of both articles establish strong pre-requisites by calibrating demographic transition dynamics that affects the economy throughout the simulation. Such an approach is also adopted in this research.

Almost all of the above research emphasizes the fiscal challenges faced by the government. One of the solutions is the transition to only partially funded pension systems. According to a widely accepted concept, pension benefits in such systems are divided into three "pillars" (Yermo, 2002). The first pillar consists of government-funded fixed pensions, typically financed through income taxes, designed to ensure a minimum standard of living for all, including the most vulnerable populations. The second pillar involves pension schemes managed by employees under employment contracts. As per the World Bank's framework, this pillar also includes the establishment of individual savings accounts (Holzmann, 2005). The third pillar comprises private savings held in special retirement accounts, aimed primarily at smoothing income for wealthier individuals.

The OLG modelling of the transition to such a multi-pillar system is also the topic of many thorough research papers, for example (Wolf & Caridad y Ocerin, 2021). Following this conceptualization and the above article approach, this paper also aims at simulating the transition to two-pillar pension system in Russia, with third pillar of private savings being the dominant one.

The OLG models are also known for their ability to capture and simulate the effects of shifting demography on the economics. Throughout the years this was one of the main reasons for development of this model class. For example, in (Kitao, 2015) OLG modelling is used to quantify the unprecedented fiscal challenges faced by Japan as it experiences the world's most rapid demographic transition. Author approach builds upon the life-cycle models that incorporate endogenous labour supply decisions in both intensive and extensive margins, which distinguishes the study from earlier work that assumed exogenous labour supply. Following such an approach this study also incorporates endogenous individual labour supply decisions.

Section 2 formulates the model, the demography transition dynamics is constructed, the function of household utility from consumption and labour is defined, and a budget constraint is set. Then the Cobb–Douglas production function with exogenous technological growth is considered. Market clearing conditions are outlined. Later, dynamic Euler's equations are derived, the model is fully solved and equilibrium is found. In the end a numerical algorithm to calculate the steady state equilibrium is given. In Section 3 the calibration is presented. The forecast of population distribution is taken from the United Nations (UN) projections, while the immigration rates are approximated using a cubic spline. Leveraging the constraint least squares method we estimate parameters of the Cobb–Douglas production function. Parameters for the utility function of consumption are selected from the literature; the generalized method of moments is used to obtain parameters for the utility of labour. Institutional parameters are set based on theory, and motivation is given to their choice. Next, the quality of fit of the calibrated model is considered. In Section 4 the expected results are summarized.

2 The model

The model used is an OLG model, with individuals living for at maximum S periods, dynamic demography, with perfect expectations, endogenous labour supply and the public tax sector. It is based on a model from (Evans & DeBacker, 2017).

2.1 Demography

Some number $\omega_{s,t}$ of households of age s are alive at period t. A measure $\omega_{1,t}$ of individuals is born each period, they are unemployed and not market-active until age E due to being too young. After age E they join the broad economic activity and they live for up to S years.

The population movement is modelled following a popular approach in demographic literature (Raftery & Ševčíková, 2023),

$$
\begin{aligned}
\omega_{1,t+1} &= (1-\rho_{0,t})\sum_{s=1}^{S} f_s \omega_{s,t} + i_1 I_t; \quad \forall t \\
\omega_{s+1,t+1} &= (1-\rho_{s,t})\omega_{s,t} + i_{s+1} I_t; \quad \forall t, \quad 1 \le s \le S-1
\end{aligned}
\tag{2.1}
$$

where $\omega_{s,t}$ is population of age s at period t, $\rho_{s,t}$ is a probability of death of age s individual at period t and f_s is an age-specific fertility rate. Net migration consists of migration rate of age s individuals and of total net-migration I_t at period t.

Such demography system setup allows to take forecasts of $\omega_{s,t}$, $\rho_{s,t}$ and I_t from advanced forecasting models, like a Bayesian probabilistic projections (Raftery, Li, Ševčíková, Gerland, & Heilig, 2012), while maintaining the structural integrity of the OLG model built on top of this demography system.

We also define the total population N_t at time t and the growth rate of the population $g_{n,t}$ at time t (2.2),

$$
\begin{aligned}
N_t &= \sum_{s=1}^{S} \omega_{s,t} \\
g_{n,t+1} &= \frac{N_{t+1}}{N_t} - 1
\end{aligned}
\tag{2.2}
$$

And additionally we define $\tilde{N}_t = \sum_{s=E+1}^{S} \omega_{s,t}$ and $\tilde{g}_{n,t+1} = \tilde{N}_{t+1}/\tilde{N}_t - 1$, which are the total working population and the working population growth rates respectively.

2.2 Households

Some number $\omega_{1,t}$ of identical individuals are born each period t and they live for at most S years, where s is their age. After reaching a working age E an individual becomes economically active, and in each period is endowed with a unit measure of time $\tilde{l}$ which he can spend either on labour or on leisure.

Each period of their lives individuals choose their sets of consumption $\{c_{s,t+s-E-1}\}_{s=E+1}^{S}$, labour $\{n_{s,t+s-E-1}\}_{s=E+1}^{S}$ and savings $\{b_{s+1,t+s-E}\}_{s=E+1}^{S-1}$ so as to maximize a lifetime-utility function,

$$\max_{\substack{\{c_{s,t+s-E-1},n_{s,t+s-E-1}\}_{s=E+1}^{S};\\ \{b_{s+1,t+s-E}\}_{s=E+1}^{S-1}}} \sum_{s=E+1}^{S} \beta^{s-1} \left[\Pi_{u=E}^{s-1}(1-\rho_{u,t+s-E-1})\right] u(c_{s,t+s-E-1}, n_{s,t+s-E-1}) \tag{2.3}$$

where β is a subjective discounting factor. Agents account for their mortality probability when choosing consumption for their future selves, thus the probability of survival term $(1-\rho_{u,t+s-E-1})$ is included in the product to represent a cumulative probability of survival. The above equation is a composition of instantaneous utility functions,

$$u(c_{s,t}, n_{s,t}) = \frac{(c_{s,t})^{1-\sigma}}{1-\sigma} + e^{tg_y(1-\sigma)} b \left[1 - \left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon}\right]^{\frac{1}{\upsilon}}; \quad \forall\, t;\; s > E \tag{2.4}$$

where σ is a relative risk aversion factor, parameters b and υ are from elliptical disutility of labour function, and $e^{tg_y(1-\sigma)}$ is added to the function for the purposes of stationarization, more on which in subsection 2.5.

Each period individual maximize their utility (2.4) while being subject to a budget constraint:

$$\begin{aligned} c_{s,t} + b_{s+1,t+1} &= (1-\tau^{l}_{s,t}) w_t n_{s,t} + (1 + r_t(1-\tau^{k}_{s,t})) b_{s,t} + X_{s,t} + \frac{BQ_t}{\tilde{N}_t} \\ BQ_t &= (1+r_t) \sum_{s=E+2}^{S} \rho_{s-1,t-1}\omega_{s-1,t-1} b_{s,t}; \quad b_{E+1,t} = b_{E+S+1,t} = 0 \end{aligned} \tag{2.5}$$

where $b_{s+1,t+1}$ are the savings of s-th individual in period t which will return to him in the next period with an interest rate of r_t; $b_{s,t}$ are the savings with which individual s enters the period t. Current wages are w_t and are equal for all individuals. Taxes $\tau^{l}_{s,t}$ and $\tau^{k}_{s,t}$ are collected, in the same period all the collected money are paid out to pensioners via the transfer component $X_{s,t}$.

Accidental bequests BQ_t are left from the households who passed away at period $t-1$ and the division by $\tilde{N}_t$ represents the fact that bequests are evenly distributed between the cohorts at period t. Agent don't have any savings from the previous period when they just entered the workforce $b_{E+1,t} = 0$, they also do not save in the last possible period of their lives since they are aware of their inevitable death next period $b_{E+S+1,t} = 0$.

2.3 Firms

There are a certain number of identical, completely competitive firms in the economy, they lease investment capital K_t from individuals at a real interest rate r_t and employ labour L_t for real wages w_t. Firms use their capital K_t and labour L_t to produce Y_t products in each period according to the Cobb–Douglas production function with exogenous technological progress g_y,

$$Y_t = AK_t^{\alpha}(e^{g_y t}L_t)^{1-\alpha} \quad \forall t, \quad \alpha \in (0,1), \quad A > 0 \tag{2.6}$$

where α is a capital share in output and A is a factor productivity. The rate of exogenous technological progress is assumed to be impacting the output exponentially. It is also assumed that a price of a unit measure of output Y_t is set $P_t = 1, \ \forall t$.

Each period companies choose K_t and L_t in order to maximize their profits given by,

$$\max \pi = \max_{K_t, L_t} \left[(1-\tau_t^c)\left(AK_t^{\alpha}(e^{g_y t}L_t)^{1-\alpha} - w_t L_t - \delta K_t\right) - r_t K_t\right] \tag{2.7}$$

where $\delta \in [0,1]$ - a capital depreciation factor and τ_t^c is a tax rate on book profits, therefore, labour costs and depreciation charges are deductible, but capital interest payments aren't.

From maximization problem in (2.7) we derive the following two equations,

$$w_t = (1-\alpha)\left[\frac{Y_t}{L_t}\right] \tag{2.8}$$

$$r_t = (1-\tau_t^c)\left[\alpha\frac{Y_t}{K_t} - \delta\right] \tag{2.9}$$

Plugging (2.6) in both (2.8) and (2.9) and then plugging (2.9) into (2.8) yields,

$$w_t = A(1-\alpha)\left[\frac{1}{A\alpha}\left(\frac{r_t}{1-\tau_t^c} + \delta\right)\right]^{\frac{\alpha}{\alpha-1}} \tag{2.10}$$

2.4 Market clearing

In this model, an equilibrium of supply and demand is established in the labour, capital, and commodity markets:

$$L_t = \sum_{s=E+1}^{S} \omega_{s,t} n_{s,t} \tag{2.11}$$

$$K_t = \sum_{s=E+2}^{S} (\omega_{s-1,t-1} b_{s,t} + i_s I_{t-1} b_{s,t}) \tag{2.12}$$

$$\begin{aligned} Y_t = C_t + Inv_t - \sum_{s=E+2}^{S} i_s I_t b_{s,t+1} \\ \text{where} \quad Inv_t \equiv K_{t+1} - (1-\delta) K_t \\ \text{and} \quad C_t \equiv \sum_{s=E+1}^{S} \omega_{s,t} c_{s,t} \end{aligned} \tag{2.13}$$

Bequests of the households who passed away in the previous period $t-1$ are formed in this period,

$$BQ_t = (1 + r_t) \sum_{s=E+2}^{S} \rho_{s-1,t-1} \omega_{s-1,t-1} b_{s,t}, \quad \forall t \tag{2.14}$$

And also pensions $X_{s,t}$ are formed and evenly paid to all retirees,

$$X_{s,t} = \frac{1}{N_{R,t}} \left[\left(\sum_{s=E+1}^{S} \omega_{s,t} (\tau^l_{s,t} w_t n_{s,t} + \tau^k_{s,t} r_t b_{s,t}) \right) + \tau^c_t (Y_t - w_t L_t - \delta K_t) \right] \tag{2.15}$$

where $N_{R,t} = \sum_{s=R+1}^{S} \omega_{s,t}$ is the total number of retirees. For pensioners the pension payment is as in (2.15), for everybody else it is obviously 0.

Since the transition period state pensions are assumed to be fixed we need variable taxes to fulfil the necessary budget income at each period, thus we have,

$$\begin{aligned} \tau^l_{s,t} &= \frac{X_t N_{R,t}}{\sum_{s=E+1}^{R} \omega_{s,t} w_t n_{s,t}}; \\ \tau^k_{s,t} &= \frac{X_t N_{R,t}}{\sum_{s=E+1}^{S} \omega_{s,t} r_t b_{s,t}}; \\ \tau^c_t &= \frac{X_t N_{R,t}}{Y_t - w_t L_t - \delta K_t} \end{aligned} \tag{2.16}$$

2.5 Stationarity

In current model setup no steady state equilibrium is possible since most of the variable are growing (i.e. having a trend component) at either the labour augmenting technological change g_y from (2.6) or at the population growth rate $g_{n,t}$ from (2.2).

The table 1 shows the sources of growth of variables in the model and their respective stationary versions. It is important to notice that r_t in (2.9) is already stationary since Y_t grows at the same rate as K_t. The individual labour supply $n_{s,t}$ is also stationary because of the $e^{tg_y(1-\sigma)}$ term in (2.4), this is demonstrated later this section.

Table 1: Stationary variable definitions

Sources of growth		
$\mathbf{e^{g_y t}}$	$\mathbf{\tilde{N}_t}$	$\mathbf{e^{g_y t}\,\tilde{N}_t}$
$\hat{c}_{s,t} \equiv \frac{c_{s,t}}{e^{g_y t}}$	$\hat{\omega}_{s,t} \equiv \frac{\omega_{s,t}}{\tilde{N}_t}$	$\hat{Y}_t \equiv \frac{Y_t}{e^{g_y t}\,\tilde{N}_t}$
$\hat{b}_{s,t} \equiv \frac{b_{s,t}}{e^{g_y t}}$	$\hat{L}_t \equiv \frac{L_t}{\tilde{N}_t}$	$\hat{K}_t \equiv \frac{K_t}{e^{g_y t}\,\tilde{N}_t}$
$\hat{w}_t \equiv \frac{w_t}{e^{g_y t}}$	$\hat{I}_t \equiv \frac{I_t}{\tilde{N}_t}$	$\hat{C}_t \equiv \frac{C_t}{e^{g_y t}\,\tilde{N}_t}$
$\hat{X}_{s,t} \equiv \frac{X_{s,t}}{e^{g_y t}}$	$\hat{N}_{R,t} \equiv \frac{N_{R,t}}{\tilde{N}_t}$	$\hat{BQ}_t \equiv \frac{BQ_t}{e^{g_y t}\,\tilde{N}_t}$

We now prove the stationarity of variables in table 1.

First, aggregate labour (2.11) is a function of population $\omega_{s,t}$, which is growing at the population growth rate, and stationary labour supply $n_{s,t}$. We can stationarize the aggregate labour market clearing condition by dividing both sides of (2.11) by the total economically relevant population $\tilde{N}_t$ in period t.

$$\hat{L}_t = \sum_{s=E+1}^{S} \hat{\omega}_{s,t} n_{s,t} \tag{2.17}$$

The aggregate capital clearing condition (2.12) and the total bequests equation (2.14) can be stationaryzed by dividing by $e^{g_y t}\tilde{N}_t$.

$$\hat{K}_t = \frac{1}{1+\tilde{g}_{n,t}} \sum_{s=E+2}^{S} \left(\hat{\omega}_{s-1,t-1}\hat{b}_{s,t} + i_s \hat{I}_{t-1}\hat{b}_{s,t}\right) \tag{2.18}$$

$$\hat{BQ}_t = \left(\frac{1+r_t}{1+\tilde{g}_{n,t}}\right) \sum_{s=E+2}^{S} \rho_{s-1,t-1}\hat{\omega}_{s-1,t-1}\hat{b}_{s,t}, \quad \forall t \tag{2.19}$$

The goods market clearing condition is rather intricate, thus it is important to configure its stationary version correctly since in the process of computing the equilibrium it will serve as an important sanity check.

$$\begin{aligned} \hat{Y}_t &= \hat{C}_t + \hat{Inv}_t - e^{g_y} \sum_{s=E+2}^{S} i_s \hat{I}_t \hat{b}_{s,t+1} \\ \text{where} \quad \hat{Inv}_t &\equiv e^{g_y}(1+\tilde{g}_{n,t+1})\hat{K}_{t+1} - (1-\delta)\hat{K}_t \\ \text{and} \quad \hat{C}_t &\equiv \sum_{s=E+1}^{S} \hat{\omega}_{s,t}\hat{c}_{s,t} \end{aligned} \tag{2.20}$$

To stationarize the aggregate output function (2.6) we divide by $e^{g_y t}\tilde{N}_t$,

$$\hat{Y}_t = A\hat{K}_t^{\alpha}\hat{L}_t^{1-\alpha} \quad \forall t, \quad \alpha \in (0,1), \quad A > 0 \tag{2.21}$$

It is important to define r_t here in terms of distribution of labour force and distribution of capital, so following (2.9) we have,

$$r_t = (1-\hat{\tau}_t^c)\left(\alpha A \left[\frac{\sum_{s=E+1}^{S} \hat{\omega}_{s,t} n_s}{(1+\tilde{g}_{n,t})^{-1}\sum_{s=E+2}^{S}\left(\hat{\omega}_{s-1,t-1}\hat{b}_{s,t} + i_s\hat{I}_{t-1}\hat{b}_{s,t}\right)}\right]^{1-\alpha} - \delta\right) \tag{2.22}$$

Equilibrium stationary wage is just,

$$\hat{w}_t = (1-\alpha)\left[\frac{\hat{Y}_t}{\hat{L}_t}\right] = A(1-\alpha)\left[\frac{1}{A\alpha}\left(\frac{r_t}{1-\hat{\tau}_t^c} + \delta\right)\right]^{\frac{\alpha}{\alpha-1}} \tag{2.23}$$

Since pensions are easily stationarized $\hat{X}_{s,t} \equiv \frac{X_{s,t}}{e^{g_y t}}$ we have the following stationary functions for various tax rates,

$$\begin{aligned} \hat{\tau}_{s,t}^l &= \frac{\hat{X}_t\hat{N}_{R,t}}{\sum_{s=E+1}^{R} \hat{\omega}_{s,t}\hat{w}_t n_{s,t}}; \\ \hat{\tau}_{s,t}^k &= \frac{\hat{X}_t\hat{N}_{R,t}}{\sum_{s=E+1}^{R} \hat{\omega}_{s,t} r_t \hat{b}_{s,t}}; \\ \hat{\tau}_{s,t}^c &= \frac{\hat{X}_t\hat{N}_{R,t}}{\hat{Y} - \hat{w}_t\hat{L}_t - \delta\hat{K}_t} \end{aligned} \tag{2.24}$$

Finally, the budget constraint (2.5) is stationarized in the following way,

$$\hat{c}_{s,t} = (1-\hat{\tau}_{s,t}^l)\hat{w}_t n_{s,t} + (1 + r_t(1-\hat{\tau}_{s,t}^k))\hat{b}_{s,t} + \hat{X}_{s,t} + \hat{BQ}_t - e^{g_y}\hat{b}_{s+1,t+1} \tag{2.25}$$

2.6 Equilibrium

To find equilibrium we must solve the maximization problem of (2.3) with substitution of utility function (2.4) and budget constraint (2.25). The resulting $2(S-E)-1$ equations for each period are the inter-temporal Euler equations characterising the equilibrium,

$$\frac{\partial u(\{\hat{n}_{s,t+s-E-1}\}_{s=E+1}^{S};\{\hat{b}_{s+1,t+s-E}\}_{s=E+1}^{S-1})}{\partial \hat{b}_{s+1,t+1}} = 0, s\in\{E+1,E+2,...,S-1\}$$
$$\Rightarrow (\hat{c}_{s,t})^{-\sigma} = e^{-\sigma g_y}\beta(1-\rho_{s,t})(1+r_{t+1}(1-\hat{\tau}^{k}_{s,t+1}))(\hat{c}_{s+1,t+1})^{-\sigma} \tag{2.26}$$

where $\hat{c}_{s,t}$ is from the budget constraint (2.25).

$$\frac{\partial u(\{n_{s,t+s-E-1}\}_{s=E+1}^{S};\{b_{s+1,t+s-E}\}_{s=E+1}^{S-1})}{\partial n_{s,t}} = 0, s\in\{E+1,E+2,...,S\}$$
$$\Rightarrow w_t(1-\hat{\tau}^{l}_{s,t})(c_{s,t})^{-\sigma} = e^{g_y t(1-\sigma)}\left(\frac{b}{\tilde{l}}\right)\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon-1}\left[1-\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon}\right]^{\frac{1-\upsilon}{\upsilon}} \Bigg| \times e^{-g_y t(1-\sigma)}$$

$$\Rightarrow \hat{w}_t(1-\hat{\tau}^{l}_{s,t})(\hat{c}_{s,t})^{-\sigma} = \left(\frac{b}{\tilde{l}}\right)\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon-1}\left[1-\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon}\right]^{\frac{1-\upsilon}{\upsilon}} \tag{2.27}$$

The equation (2.27) can be further simplified,

$$\hat{w}_t(1-\hat{\tau}^{l}_{s,t})(\hat{c}_{s,t})^{-\sigma} = \left(\frac{b}{\tilde{l}}\right)\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon-1}\left[1-\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon}\right]^{\frac{1-\upsilon}{\upsilon}}$$
$$\Rightarrow \hat{w}_t(1-\hat{\tau}^{l}_{s,t})(\hat{c}_{s,t})^{-\sigma} = \left(\frac{b}{\tilde{l}}\right)\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon-1}\left(\frac{n_{s,t}}{\tilde{l}}\right)^{1-\upsilon}\left[\left(\frac{n_{s,t}}{\tilde{l}}\right)^{-\upsilon}-1\right]^{\frac{1-\upsilon}{\upsilon}}$$
$$\Rightarrow \hat{w}_t(1-\hat{\tau}^{l}_{s,t})(\hat{c}_{s,t})^{-\sigma} = \left(\frac{b}{\tilde{l}}\right)\left[\left(\frac{n_{s,t}}{\tilde{l}}\right)^{-\upsilon}-1\right]^{\frac{1-\upsilon}{\upsilon}}$$

And thus we get,

$$n_{s,t} = \tilde{l}\left[1+\left(\frac{b(\hat{c}_{s,t})^{\sigma}}{\tilde{l}\hat{w}_t(1-\hat{\tau}^{l}_{s,t})}\right)^{\frac{\upsilon}{\upsilon-1}}\right]^{-\frac{1}{\upsilon}} \tag{2.28}$$

Now we can define the algorithm of numerically finding the steady state of the system. For the algorithm we need at minimum two guesses of variables values for the outer loop, which will allow us to solve for the steady state values of the entire system. The choice of two variable for which we will make a guess is r_t and specific $\tau_{s,t}$ since it is much easier to make an initial guess for those fraction variables as they all are

$\in (0, 1)$.

It is important to explain what is a *specific* $\tau_{s,t}$. For the purposes of this research the three possible transition scenarios are created, they differ only in the way they deal with the fiscal tax burden due to pension payments. They correspond to financing the budget gap by three different taxes, all of which use just one tax rate. Depending on the scenario and a specific tax financing the transition we choose a tax rate variable for the outer loop solution.

In what follows, steady state values of variables are indicated by a bar sign.

Algorithm 1 Steady State solution algorithm

Require: make a guess for $\bar{r}$, a specific $\bar{\tau}$ and $\bar{BQ}$

repeat

$\bar{w} \leftarrow \{\bar{r}, \bar{\tau}^c, A, \alpha, \delta\}$ according to (2.23)

▷ A rootfinding algorithm

Make a guess for $\bar{c}_{E+1}$

$\{\bar{c_s}\}_{s=E+1}^{S} \leftarrow \{\bar{c_1}, \bar{r}, \bar{\tau}_s^k, \rho_s, \beta, \sigma, g_y\}$ according to (2.26)

$\{\bar{n_s}\}_{s=E+1}^{S} \leftarrow \{\bar{w}, \bar{c_s}, \bar{\tau}_s^l, \tilde{l}, \sigma, \upsilon, b\}$ according to (2.28)

$\{\bar{b_s}\}_{s=E+2}^{S} \leftarrow \{\bar{c_s}, \bar{w}, \bar{n_s}, \bar{\tau}_s^l, \bar{r}, \bar{\tau}_s^k, \bar{X}_s, \bar{BQ}, g_y\}$ according to (2.25), $\bar{b}_{s=E} = 0$

Rootfinder optimizes initial $\bar{c}_{E+1}$ guess so as to satisfy $b_{s=S+1} = 0$ – in the last period of agent live savings are zero since he is aware of his inevitable death next year.

Then, as above, find $\{\bar{c_s}\}_{s=E+1}^{S} \rightarrow \{\bar{n_s}\}_{s=E+1}^{S} \rightarrow \{\bar{b_s}\}_{s=E+2}^{S}$

$\bar{BQ}' \leftarrow \{\bar{r}, \bar{b}_s, \bar{\omega}_s, \bar{\rho}_s, \tilde{g}_n\}$ according to (2.19)

$\bar{r}' \leftarrow \{\bar{\tau}^c, \bar{n}_s, \bar{b}_s, \bar{\omega}_s, \bar{I}, \bar{i}_s, \tilde{g}_n, \alpha, A, \delta\}$ according to (2.22)

$\bar{\tau}' \leftarrow \{\bar{n}_s, \bar{b}_s, \bar{w}, \bar{r}, \bar{X}_s, \bar{Y}, \bar{\omega}_s, \bar{N}_R, \delta\}$ according to (2.24)

Update initial guesses: $\{\bar{r};\ \bar{\tau};\ \bar{BQ}\}_i = \{\xi\bar{r}' + (1-\xi)\bar{r};\ \xi\bar{\tau}' + (1-\xi)\bar{\tau};\ \xi\bar{BQ}' + (1-\xi)\bar{BQ}\}$

until $\{(\bar{r}' - \bar{r})^2;\ (\bar{\tau}' - \bar{\tau})^2;\ (\bar{BQ}' - \bar{BQ})^2\} < \{r_{tol};\ \tau_{tol};\ BQ_{tol}\}$

When the algorithm is converged you obtain all the variables for all ages of the model in the steady state, thus the algorithm allows to fully define the steady state equilibrium.

3 Calibration

This section presents the values for all the parameters in the model, described in section 2, that are calibrated to simulate the Russian economy. It also shows the methodology of estimating parameters statistically or setting them based on theory or literature.

3.1 Demographic parameters $\{\omega_{s,t}, \rho_{s,t}, I_t, i_s\}$

As previously mentioned in subsection 2.1, the demography system setup allows to use any forecast estimates for the demographic parameters. In this case, we use forecasts for age-s number of households at year t is given by $\underset{S\times T}{\boldsymbol{\Omega}} = \{\{\omega_{s,t}\}_{s=1}^{S}\}_{t=2025}^{T=2100}$, for age-$s$ probability of death at year t is given by $\underset{S\times T}{\mathbf{P}} = \{\{\rho_{s,t}\}_{s=1}^{S}\}_{t=2025}^{T=2100}$ and for total net migration at year t is given by $\underset{1\times T}{\boldsymbol{I}} = \{I_t\}_{t=2025}^{T=2100}$, we use the most recent UN Population Projections data for Russia, since it is the best cohort-based forecast of the population of Russia (UN, Population Division, 2024). Their population forecasts are given until the year 2100.

For the net migration rates by age-s given by $\underset{S\times 1}{\boldsymbol{i}} = \{i_s\}_{s=1}^{S}$ we estimate the average such rates for age-s individuals across 7 years (2017-2023) of age specific migration data available for Russia. The data is taken from Unified Interdepartmental Information and Statistical System of Russia [3].

The quality of the forecast data can be seen on the following figures,

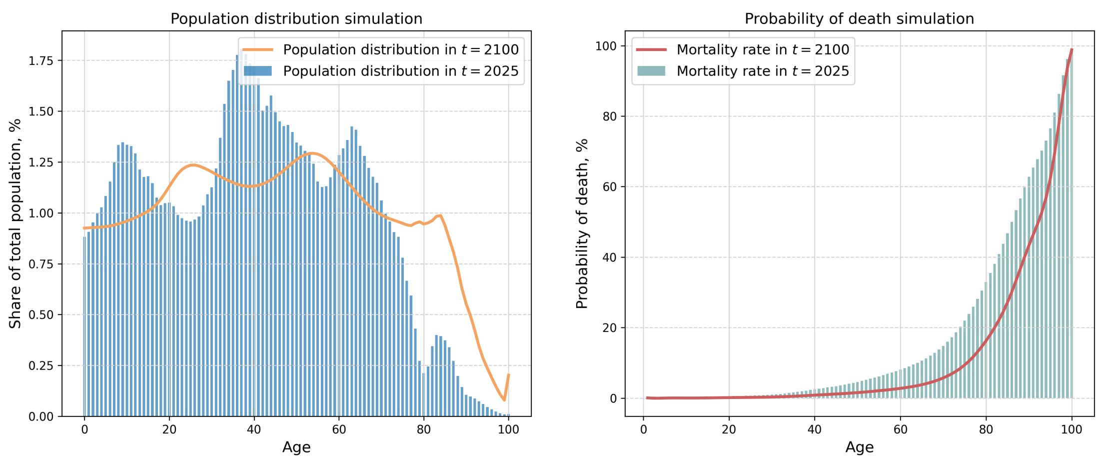

Figure 1: Population distribution and probability of death simulation

As we can see from the left figure 1, the population in year 2100 will be significantly older than today. In fact, it will also be smaller, about 130 million people. The

[3]URL: https://www.fedstat.ru/indicator/58615

number of newborn and young will decrease dramatically, while very old-age people (aged 80 and more) will drastically grow in numbers. The right part of the figure 1 demonstrates why the population will grow so old, its due to the fact that the survival probability for the aged will grow significantly to the year 2100. And while improving the life expectancy, the increased probability of survival together with little births means shrinking workforce and growing amount of elderly.

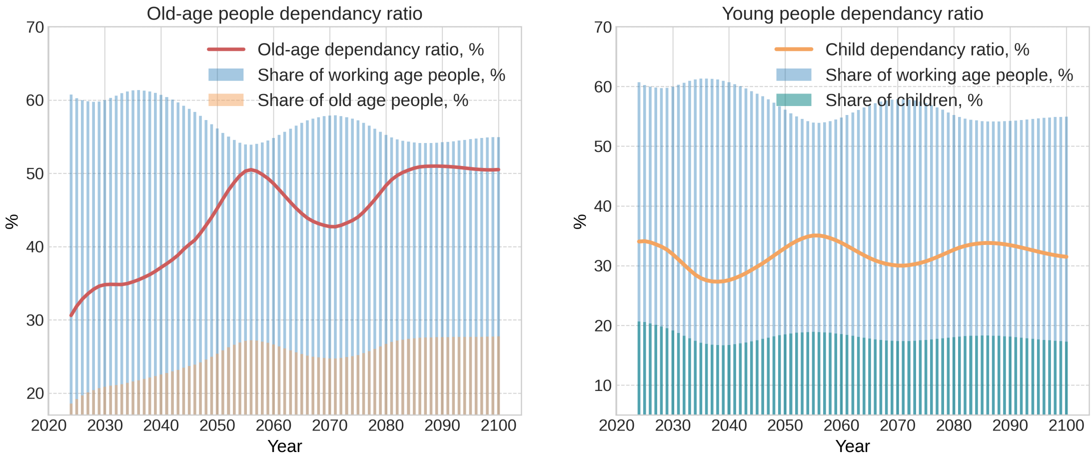

Figure 2: Old-age and young-age dependency ratios simulation

Figure 2 demonstrates the time-path of the old-age dependency ratio on the left, which is measured as the number of elderly (65+) divided by the number of working age people (18-64). As we can see, trends outlined in figure 1 are also seen here – number of young almost steadily decreases, number of elderly, on the other hand, grows. As a result, the old-age dependency ratio grows from 30% today to 50% in just 30 years. And though in later decades the ratio slightly decreases, it later reaches a new equilibrium at exactly the 50%.

The consequences of this trend in the long term are appalling: the PAYGO pension system, currently established in Russia, will no longer be able to sustain the wellbeing of retirees.

On the right of the figure 2 we see a young-age dependency ratio calculated as the number of young (0-18) to the number of working (18-65). The share of children shows similar dynamics, in relative terms, to the share of the working age people, because of that the young dependency ratio is expected to stay almost the same throughout the century.

3.2 Production function parameters $\{A, \alpha, g_y\}$

In the Cobb–Douglas production function (2.6), there is a restriction on the parameter $\alpha \in (0, 1)$, since we need $\alpha + (1 - \alpha) = 1$. In order to correctly estimate the parameters of the production function (2.6), it is necessary to carry out the following transformations:

$$\begin{gathered} Y_t = AK_t^{\alpha}(e^{g_y t}L_t)^{1-\alpha} \\ \Rightarrow \ln Y_t = \ln A + \alpha \ln (K_t) + (1-\alpha) \ln (e^{g_y t} L_t) \end{gathered}$$

$$\Rightarrow \ln Y_t = \ln A + \alpha \ln (K_t) + (1-\alpha) g_y t + (1-\alpha) \ln L_t \tag{3.1}$$

And from the t-period equation (??) subtract the same equation but in period $t-1$ to get,

$$\Delta \ln Y_t = \alpha \Delta \ln K_t + (1-\alpha)\Delta \ln L_t + (1-\alpha) g_y$$

Or equivalently we can write it in a regression-ready form,

$$\Delta \ln Y_t = \beta_0 + \beta_1 \Delta \ln K_t + \beta_2 \Delta \ln L_t + \varepsilon_t \quad \forall t;\ \beta_1 + \beta_2 = 1 \tag{3.2}$$

where $\beta_0 = (1-\alpha)g_y$, $\beta_1 = \alpha$ and $\beta_2 = (1-\alpha)$

Using constraint least squares with $\beta_1 + \beta_2 = 1$ constraint we can estimate the parameters of the (??). And then, to obtain the production function parameters we calculate,

$$\hat{\alpha} = \hat{\beta}_1; \quad \hat{g}_y = \frac{\hat{\beta}_0}{\hat{\beta}_2} \tag{3.3}$$

Now we extract A from the (??) with estimates from (??) substituted inside,

$$\ln A = \ln Y_t - \hat{\alpha} \ln (K_t) - (1-\hat{\alpha})\hat{g}_y t - (1-\hat{\alpha}) \ln L_t$$

$$\Rightarrow \hat{A} = \frac{1}{T}\sum_{t=1}^{T} \exp\left(\ln Y_t - \hat{\alpha} \ln (K_t) - (1-\hat{\alpha})\hat{g}_y t - (1-\hat{\alpha}) \ln L_t\right) \tag{3.4}$$

where $\hat{A}$ is an estimate from constant regression.

To construct regressions (??) and (??), the following data was selected for the period from 1999 to 2024:

1. Gross Domestic Product in 2024 prices (trln. of rub) as a variable for aggregate output Y_t;

2. Gross Fixed Capital Formation in 2024 prices (trln. of rub) as a variable of aggregate capital K_t;

3. The number of people employed (mln. people) as a variable of the aggregated labour supply L_t.

Estimation by constraint least squares method of (??) gave the estimates $\hat{\alpha} = 0.39$ and $\hat{g}_y = 0.054$. Estimation of (??) gave the estimate $\hat{A} = 1.403$.

The obtained estimates allow to calculate the model-implied output Y_t according to (2.6), interest rate r_t according to (2.9) and wage w_t according to (2.8). What follows are the graphs with simulated and real values,

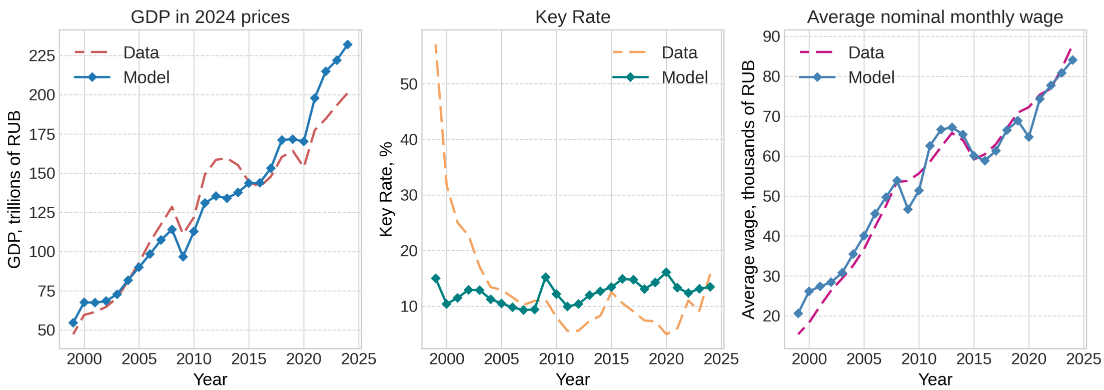

Figure 3: Simulated and real values of output, key rate, wages

As we can see from figure 3 the quality of fit of GDP and Average wage is very high and leaves no doubt of the model capability with regards to aggregate variables. The key rate dynamics, on the other hand, is far from ideal, but nonetheless, simulated interest rate is co-aligned with the real one on the most of its time-path.

3.3 Selection of parameters $\{\delta, \theta, \sigma, \beta\}$

Aside for a tax rate, the last undefined parameter of the profit maximization problem for representative firms (2.7), on which the key rate depends, is the capital depreciation coefficient δ. Following (Voskoboynikov, 2012), it was set equal to $\delta = 0.05$. There are very few macro estimates of this parameter for the Russian economy in the literature. This estimate, despite the fact that it was made more than 10 years ago, turned out to be the most recent. The intertemporal elasticity of substitution of consumption is $\sigma = 1.97$ (Shulgin, 2018); (Charnavoki, 2019). The subjective discount factor is $\beta = 0.905$ (Khvostova, Larin, & Novak, 2014).

Furthermore, for the purposes of estimating parameters in elliptical disutility of labour in (2.4) we need to introduce a more broadly used utility function of labour, find estimates of its parameters in the literature and then match them to our elliptical function using Generalized Method of Moments.

The most popular function for this purposes is the Frisch elasticity of labour supply function (Frisch, 1959),

$$u(n_{s,t}) = -\frac{\theta\left(n_{s,t}^{\frac{1+\theta}{\theta}}\right)}{1+\theta} \tag{3.5}$$

The estimate of Frisch constant elasticity of labour supply θ from (??) is assumed to be equal to $\theta = \frac{1}{1.77}$, following the work (Shulgin, 2018), where it was estimated on microdata. The parameters for the elliptical utility (2.4) are estimated in the next subsection 3.4

3.4 Elliptic utility parameters $\{b, \upsilon\}$

To estimate the parameters b and υ of the elliptic disutility of labour function from (2.4), it is necessary that it approximates the Frisch elasticity of labour supply function (??), where $\theta = 1/1.77 = 0.565$.

Estimates of b and υ were obtained using the generalized method of moments, where the quadratic norm of marginal utility of labour (??) was chosen as the moment:

$$L^2 = \|MU_{cfe} - MU_{ellip}\| =$$
$$= \sum\left[n_{s,t}^{\frac{1}{\theta}} - \left(\frac{b}{\tilde{l}}\right)\left(\frac{n_{s,t}}{\tilde{l}}\right)\left(1-\left(\frac{n_{s,t}}{\tilde{l}}\right)^{\upsilon}\right)^{\frac{1-\upsilon}{\upsilon}}\right]^2 \tag{3.6}$$

Minimizing (??) within $n_{s,t} \in [10^{-10}, 1+10^{-10}]$, by selecting parameters b and υ using the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (LBFGS) algorithm (Zhu, Byrd, Lu, & Nocedal, 1997), we obtain the estimates of $b = 0.431$ and $\upsilon = 1.765$.

The marginal effect of the elliptical disutility of labour function approximates the marginal effect of the Frisch elasticity of labour utility function quite accurately, as can be seen from the figure 4.

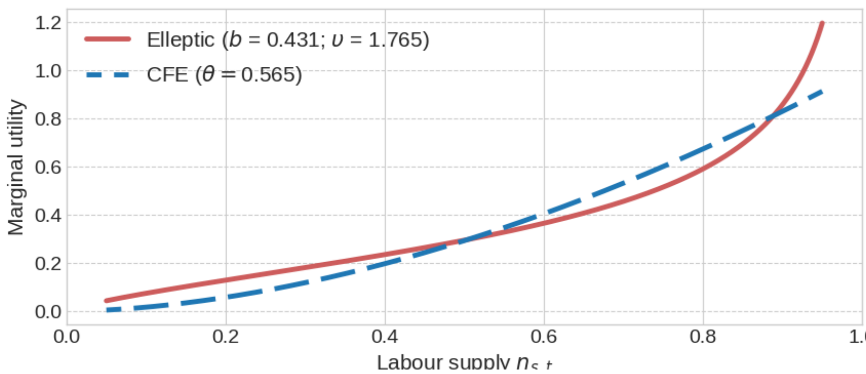

Figure 4: The marginal utility of labour for Frisch and elliptic function

3.5 Theoretical justification of parameters $\{\tilde{l}, S, E, R, \tau\}$

The time allocated by each individual for work and rest, $\tilde{l} = l_{s,t} + n_{s,t} \;\; \forall s, t$ is set to $\tilde{l} = 1$, in this way time spent on work or rests represents the shares of overall time. The number of cohorts of the population (maximum age in the model) S is set to be equal $S = 100$ since that is a usual assumptions not only on OLG modelling, but also in statistical reports that collect age-specific data for ages up to 100.

The age at which individuals start to be economically active (work, save, consume) E is set to be $E = 20$ since people aged younger have almost no participation in the economic activities, which follows from the statistical compendium (Laykam, Gimpelson, Jihareva, Zaynullina, & Kulayeva, 2018).

The retirement age will be taken as a gender average, since there is no distinction of individuals between men and women within the framework of the model. Starting in 2028, when the transition period to the new retirement age ends, the retirement ages will be 65 and 60 years for men and women, respectively. Then, the retirement age is $R = \frac{60+65}{2} = 62$.

The model does not take into account the early retirement, as the proportion of pensioners receiving pensions before reaching retirement age is decreasing. In (Derkach, 2015) it is shown that, starting from 2010 and up to 2014, the number of early retirees decreased from 15.4% to 13% of the total number of pensioners, and this trend only continued since. Thus, in 2023, the total number of retired workers amounted to 41.7 million people, of which only 4.5 million people have not reached the general retirement age, which is 10.7% of retirees [4]. Since the trend of reduction in this share is likely to continue in the future, it is not necessary to take into account early retirees within the framework of this long-term model.

The income tax, that is fully spent on paying pension, has a rate $\tau^l = 0.22$, since that is exactly how much any employer deducts to the pension fund from the salary of each its employee [5]. The rates of pension corporate profit tax $t^c = 0$ and pension tax on interest earnings $\tau^k = 0$, since these taxes are not used to pay pensions in the initial period of the model (i.e. today).

In reality, the Pension Fund receives about 2.8 trillion roubles out of the 8.8 trillion roubles spent on old-age insurance pensions from sources unrelated to the income tax. The Pension Fund's inter-budget transfers for the pension payments amounts to about 1 trillion roubles, the rest of the necessary funds (approximately 1.8 trillion roubles) is received by the Pension Fund from its other income and non-targeted budget

[4] Information on the number of pensioners registered in the SFR system who receive pensions in accordance with the Federal Law "On Insurance Pensions" and who have not reached the generally established retirement age (including those who are employed and those who are not). URL: https://sfr.gov.ru/info/statistics/pension_provision

[5] In accordance with Article 425 of the Tax Code of the Russian Federation.

transfers [6]. For the modelling purposes, this co-financing of the Pension Fund budget is assumed to be exogenous and is not taken into account in the model.

The final calibration of all the parameters is presented in the following table 2,

Table 2: Calibrated model parameters

Parameter	Description	Value
A	Total factor productivity	1.403
α	Capital share in output	0.39
g_y	Rate of labour augmenting technological growth	0.054
δ	Per-period depreciation rate of capital	0.05
σ	Coefficient of relative risk aversion	1.97
β	Per-period subjective discount factor	0.905
b	Elliptical disutility of labour scale parameter	0.431
υ	Elliptical disutility of labour shape parameter	1.765
$\tilde{l}$	Time endowment per period	1
S	Maximum number of periods in households life	100
E	Economic activity beginning age	20
R	State pension age	62
τ^l	Individual income pension tax rate	0.22
τ^k	Interest returns pension tax rate	0
τ^c	Corporate profit pension tax rate	0
$\boldsymbol{\Omega}$	Size $S \times T$ matrix of population by age and time	see 3.1
$\mathbf{P}$	Size $S \times T$ matrix of death probabilities by age and time	see 3.1
$\boldsymbol{I}$	Size $1 \times T$ vector of total net migration by time	see 3.1
$\boldsymbol{i}$	Size $S \times 1$ vector of net migration rates by age	see 3.1

[6]The structure of income and expenses of the budget of the Pension Fund of the Russian Federation for 2022. URL: https://base.garant.ru/407425284/

4 Expected Results

The model is expected to simulate three distinct fiscal approaches to financing the pension system transition deficit. The approaches correspond to three transition scenarios:

1. **Labour Tax:** Deficit financed by maintaining pre-reform labour taxes
2. **Profit Tax:** Deficit financed by corporate profit taxes
3. **Savings Tax:** Deficit financed by taxes on capital returns (savings/interest)

Firstly, we expect the following results with regards to macroeconomic variables:

1. **Aggregate Output (GDP):**
   - An initial decline in GDP is expected under the Profit Tax and Savings Tax scenarios immediately following the 2025 reform, attributed to negative shocks from shifting tax burdens onto capital.
   - A slight increase in GDP is expected under the Labour Tax scenario in the long run.
   - The dynamics of aggregate capital accumulation are predicted to be the primary driver of these output changes, outweighing labour supply effects.
2. **Aggregate Capital:**
   - Significant declines in aggregate capital are expected under both the Profit Tax and Savings Tax scenarios. This stems from the increased tax burden reducing the attractiveness and returns on savings/investment.
   - The Savings Tax scenario is expected to cause the steepest capital decline initially.
   - Growth in aggregate capital is expected under the Labour Tax scenario, as individuals increase private savings to prepare for retirement without state pensions.
3. **Aggregate Labour Supply:**
   - Increases are expected under the Profit Tax and Savings Tax scenarios. This arises because:
     (a) Reduced post-tax returns on savings make labour relatively more attractive.

(b) Cohorts nearing retirement who saved sub-optimally need to work more to accumulate sufficient retirement funds.

- A moderate decrease is expected under the Labour Tax scenario, driven by the negative impact of higher labour taxes on the marginal return to work.

Secondly, the effect on aggregate and age-group specific consumption is expected to be the following:

1. **Working-Age Individuals:**
   - Increased average consumption is expected across all scenarios post-reform due to lower taxes and higher disposable income.
   - The strongest consumption growth is expected under the Savings Tax scenario, where the relative attractiveness of saving is most diminished.

2. **Pensioners:**
   - A significant initial decline in consumption is expected during the early transition phase. This results from:
     (a) Reduced state pension payments (to minimum wage levels).
     (b) The first retiring cohorts being individuals who saved sub-optimally during their working lives.
   - A subsequent increase in average pensioner consumption is expected later in the transition. This is driven by retiring cohorts who saved optimally throughout their lives and accumulated sufficient private funds, even as state pension recipients dwindle.

While the model is expected to generate interpretable and broadly realistic results illustrating the economy's dynamics during the pension transition, the findings highlight the significant trade-offs between different financing mechanisms, particularly concerning capital accumulation, labour supply incentives, and intergenerational equity in consumption.

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Appendix

A Data sources

The original data sources are listed in the paper itself. The resulting data combined and the Python code are available at
https://drive.google.com/drive/folders/1YnJRj94y4wHgxWUQuH4hA_fGYL2W0b_3?usp=sharing